Robotics and Flexible Automation

Lecture 3
DH modelling and forward kinematics

Prof. Zengxi (Stephen) Pan

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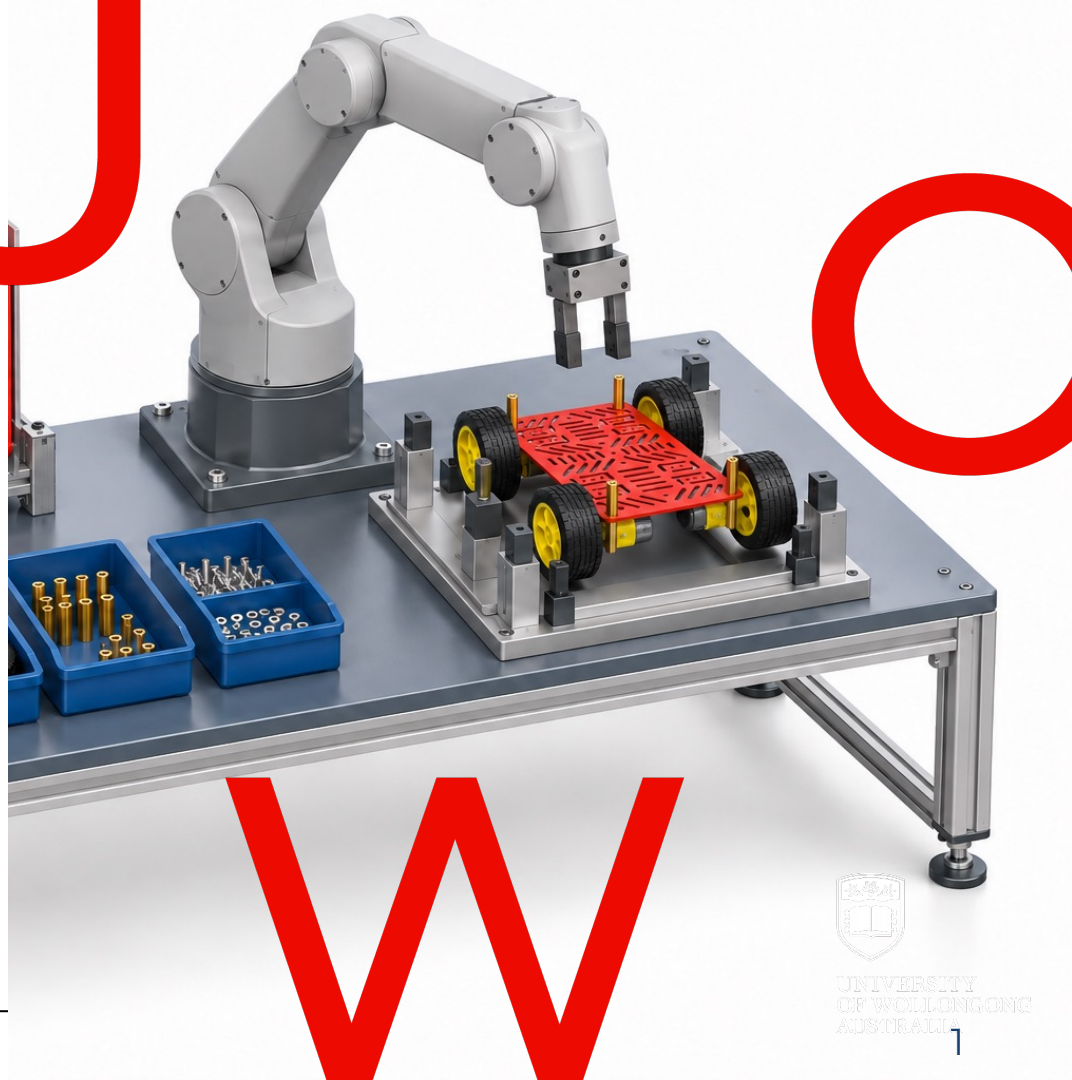


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3 From DH table to arm equation

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PART 1

DH modelling

A repeatable method for assigning frames to a robot

1. Kinematic modelling and link frames
2. Standard DH frame assignment
3. The four DH parameters and edge cases

Forward and Inverse Kinematics

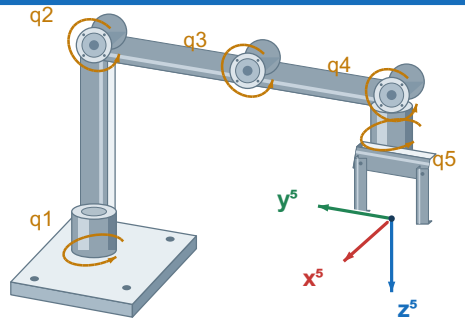
CORE

Forward kinematics (this class)

Joint coordinates

$$q = [q_1 \ \dots \ q_n]^T$$

Joint angles / displacements



Tool pose:
Cartesian coordinates

$$T$$

Position and orientation

Inverse kinematics (next class)

Forward: a joint configuration determines one tool pose.
Inverse: a desired pose may have no solution, one solution or multiple solutions.

Kinematic Modelling: History and Approaches

CORE

Earlier foundations **Rigid-body motion**	Describe rigid motion using rotations and translations. Screw theory combines rotation and translation along an axis.
1955 **Standard DH**	Denavit–Hartenberg: four parameters per adjacent link pair. A systematic frame convention for a serial kinematic chain.
Later variant **Modified DH**	Different frame placement and transformation order. Both conventions work, but their tables and formulas differ.
Product of Exponentials **PoE formulation**	PoE uses joint screw axes and a reference pose to calculate T(q). It avoids assigning a DH frame to every link.
Modern ROS workflow **URDF**	An XML robot model: links, joint axes and relative origins. Software combines this model with q to compute link poses.

Frames Fixed to Links

CORE

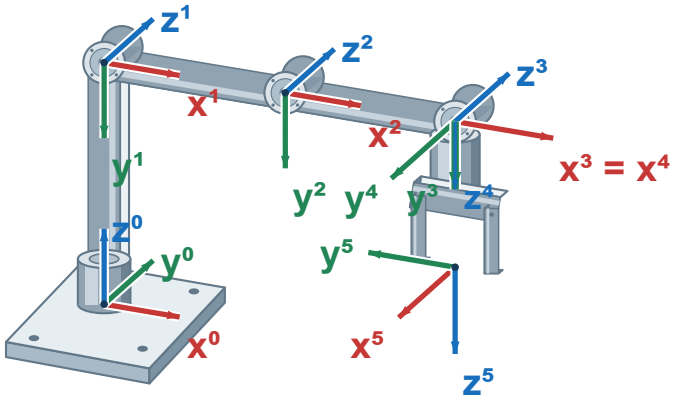
Each connecting joint determines one neighbouring-frame relationship

One frame is fixed to each link

The frame moves with its link. Its position and orientation relative to that link stay constant.

One joint sets the adjacent pose

With the link geometry fixed, A_i depends only on the connecting joint coordinate q_i.



$$A_i(q_i) = {}^{i-1}T_i$$

Each joint variable q_i appears only in the neighbouring transform $A_i(q_i)$. The complete pose is the ordered product ${}^0T_n = A_1(q_1)\,A_2(q_2)\cdots A_n(q_n)$.

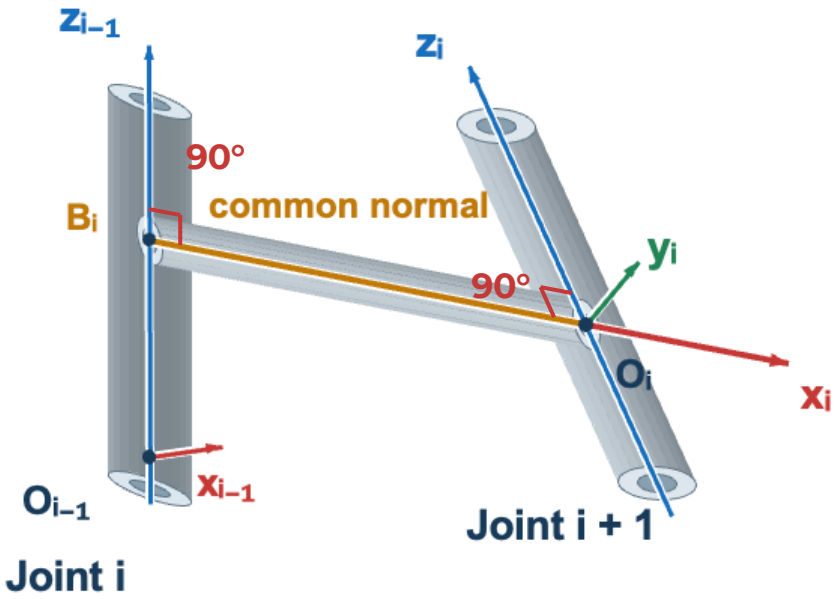
Standard DH Frame Assignment

CORE

Five rules, applied joint by joint from the base

1. Number the joints 1 … n from the base.
2. Align z(i−1) with the axis of joint i.
3. Choose x(i) along the common normal.
4. Put the origin O(i) where x(i) meets z(i).
5. Complete a right-handed frame.

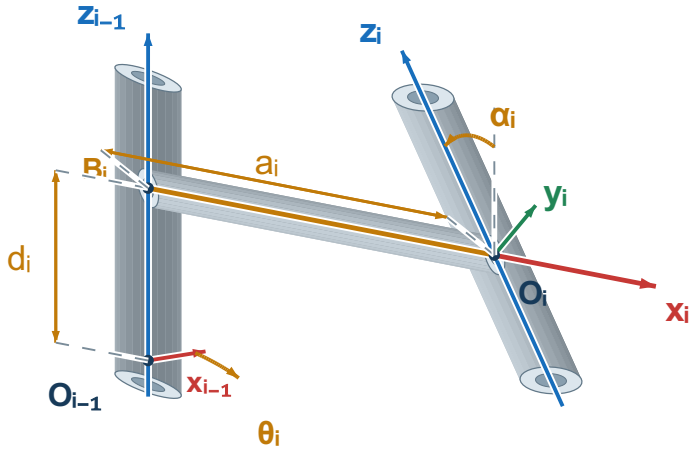
$$\hat{y}_i = \hat{z}_i \times \hat{x}_i$$



The Four DH Parameters

Two describe the joint, two describe the link

Parameter	How it is measured
θi	rotation from x(i−1) to x(i), about z(i−1)
di	signed displacement along z(i−1)
ai	signed displacement along x(i)
αi	rotation from z(i−1) to z(i), about x(i)



Revolute: θ_i varies with q_i.

Prismatic: d_i varies with q_i.

Home is the reference position used to define zero joint motion.
The Home column records θ_i (revolute) or d_i (prismatic) at that position.

D-H Algorithm I: Attach Frames

CORE

5 joints need 6 frames: {0}, {1}, {2}, {3}, {4}, {5}

0. Number the joints 1 to n from the base to the tool.
1. Attach L_0 to the base. Align **z_0 with joint 1**. Set k = 1.
2. Align **z_k with the axis of joint k + 1**.
3. Place O_k where z_k meets the common normal to z_{k-1}. If the axes intersect, use their intersection.
4. Choose x_k perpendicular to both z axes. For separated axes, use their common normal.
5. Choose $y_k = z_k \times x_k$ to complete a right-handed frame.
6. Set k = k + 1. If **k < n**, return to step 2.
7. Attach **L_n to the tool**. Set **z_n along approach** and y_n along gripper opening; complete x_n.

D-H Algorithm II: Kinematic Parameters

CORE

Read the four parameters between neighbouring link frames

8 Locate b_k where the common normal x_k meets the preceding joint axis z_{k-1}.

9 θ_k: rotation from x_{k-1} to x_k, measured **about z_{k-1}**.

10 d_k: signed distance from O_{k-1} to b_k, measured **along z_{k-1}**.

11 a_k: signed distance from b_k to O_k, measured **along x_k**.

12 α_k: rotation from z_{k-1} to z_k, measured **about x_k**.

13 Set k = k + 1. If **$k \le n$**, return to step 8; otherwise stop.

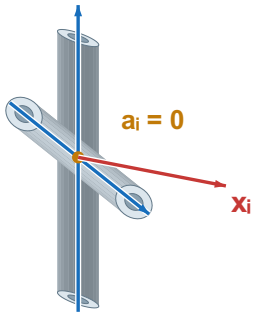
For each joint, fill one row of the DH table: θ_k d_k a_k α_k

Intersecting, Parallel and Collinear Axes

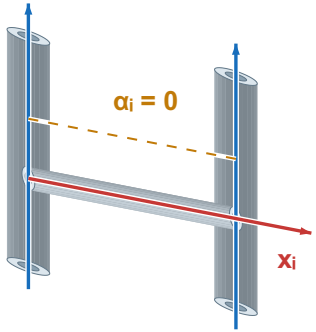
CORE

Apply steps 3–4 in D-H Algorithm I (page 9): choose the origin and common normal for each case

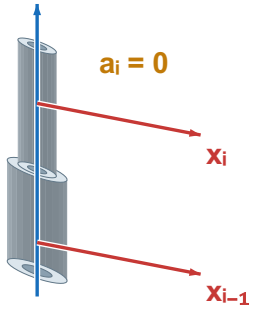
Intersecting axes



Parallel axes



Collinear axes



Intersecting axes — the common normal has zero length, so a(i) = 0.

Parallel axes — the position along the axis is free; pick one and keep it.

Collinear axes — origin and transverse axes are free; stay consistent.

PART 2

Worked robot models

Build a DH table from physical robot geometry

1. Alpha II 5-axis robot
2. Adept One SCARA example

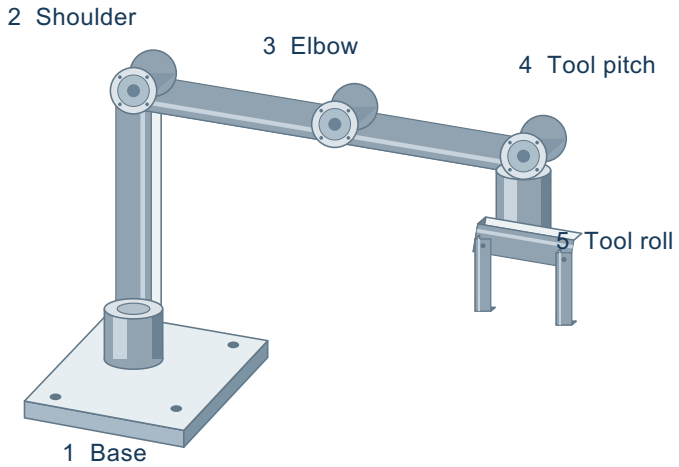
Example 1: Alpha II, 5-Axis Robot

EXAMPLE

Microbot Alpha II: physical robot



Five-joint kinematic schematic

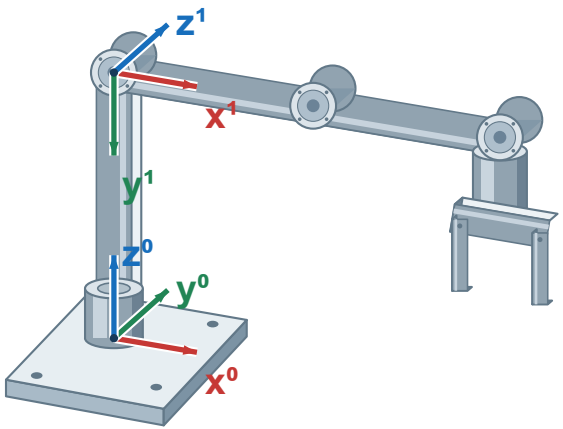


5 revolute joints: frames {0} to {5}, giving 6 frames and 5 DH rows.

Example 1: Alpha II, Frame 1 and DH Row 1

EXAMPLE

Locate the axes and origin first, then read the four DH parameters



Start with z_0 along the vertical base axis. Place O_0 on that axis.

Joint 2 fixes z_1. Its axis intersects z_0 at the shoulder, so O_1 is at that intersection.

Choose x_1 along the arm in the reference pose. Complete y_1 by the right-hand rule.

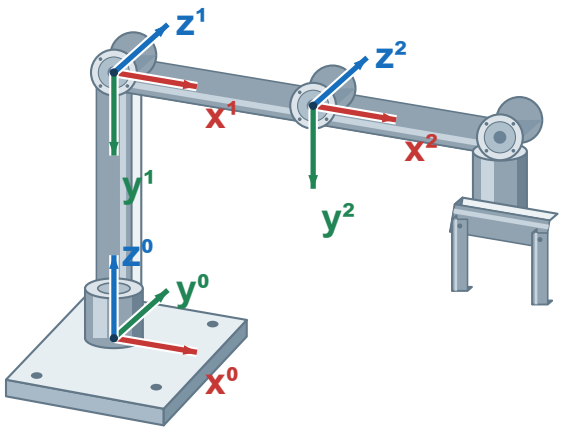
Read the geometry: d_1 = 215 mm, a_1 = 0 and α_1 = −90°. Only θ_1 varies.

Axis	θ	d	a	α	Home
1	*ϑ1*	*d1*	*0*	*−90°*	*0°*

Example 1: Alpha II, Frame 2 and DH Row 2

EXAMPLE

Parallel joint axes: the common normal follows the upper arm



Frame 2 belongs to link 2. Align z_2 with joint 3 at the elbow.

z_1 and z_2 are parallel. Their common normal follows the upper arm.

Place O_2 at the elbow and point x_2 along the common normal from z_1 to z_2.

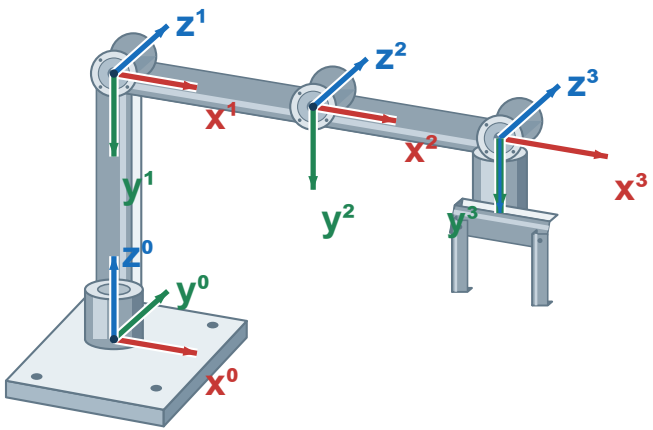
The upper-arm length gives a_2 = 177.8 mm. $d_2 = 0$, $\alpha_2 = 0°$; θ_2 varies.

Axis	θ	d	a	α	Home
2	*ϑ2*	*0*	*a2*	*0°*	*0°*

Example 1: Alpha II, Frame 3 and DH Row 3

EXAMPLE

Parallel joint axes: the common normal follows the forearm



Frame 3 belongs to link 3. Align z_3 with joint 4 at the wrist.

z_2 and z_3 are parallel. Their common normal follows the forearm.

Place O_3 at the wrist and choose x_3 along the forearm from z_2 to z_3.

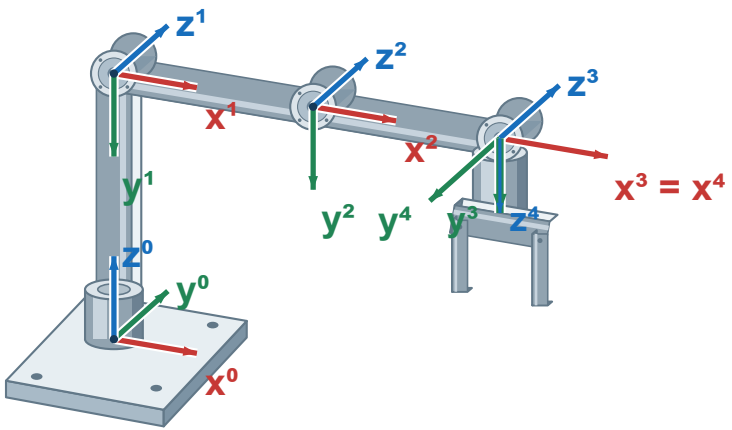
The forearm length gives a_3 = 177.8 mm. d_3 = 0, α_3 = 0°; θ_3 varies.

Axis	θ	d	a	α	Home
3	*ϑ3*	*0*	*a3*	*0°*	*0°*

Example 1: Alpha II, Frame 4 and DH Row 4

EXAMPLE

Intersecting wrist axes: two link frames share one origin



Frame 4 belongs to link 4. Align z_4 with joint 5, the tool-roll axis.

z_3 and z_4 intersect at the wrist. Choose $O_4 = O_3$, giving $a_4 = d_4 = 0$.

Choose $x_4 = x_3$ in the reference pose. Complete the right-handed frame.

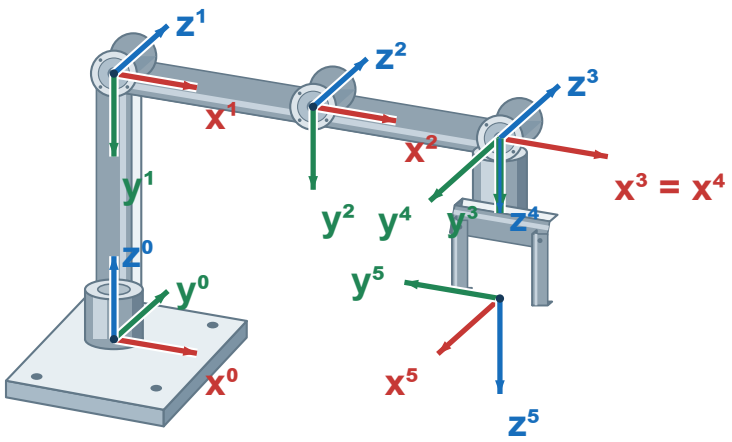
The twist from z_3 to z_4 about x_4 is $\alpha_4 = -90°$. Only θ_4 varies.

Axis	θ	d	a	α	Home
4	$\vartheta 4$	0	0	−90°	0°

Example 1: Alpha II, Frame 5 and DH Row 5

EXAMPLE

Final frame: tool roll, the tool offset and the reference angle



Frame 5 belongs to the final link. Align z_5 with the tool approach axis, parallel to z_4.

Place O_5 a distance d_5 = 129.5 mm along z_4 from O_4.

Choose x_5 and y_5 with the gripper orientation. In the drawn pose, $\theta_5 = 90°$.

Read $a_5 = 0$ and $\alpha_5 = 0°$. θ_5 varies. A different TCP requires a fixed tool transform.

Axis	θ	d	a	α	Home
5	*ϑ5*	*d5*	*0*	*0°*	*90°*

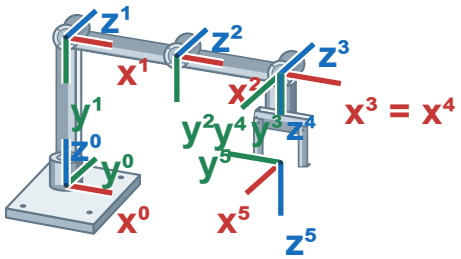
Example 1: Alpha II, Complete DH Table

EXAMPLE

Combine the five rows derived from frames 0–5

Axis	θ	d	a	α	Home
1	*ϑ1*	*d1*	*0*	*−90°*	*0°*
2	*ϑ2*	*0*	*a2*	*0°*	*0°*
3	*ϑ3*	*0*	*a3*	*0°*	*0°*
4	*ϑ4*	*0*	*0*	*−90°*	*0°*
5	*ϑ5*	*d5*	*0*	*0°*	*90°*

$d_1 = 215$ mm $a_2 = a_3 = 177.8$ mm $d_5 = 129.5$ mm



Example 1: Alpha II, Zero and Reference Positions

EXAMPLE

Five joint coordinates describe the configuration of this same robot

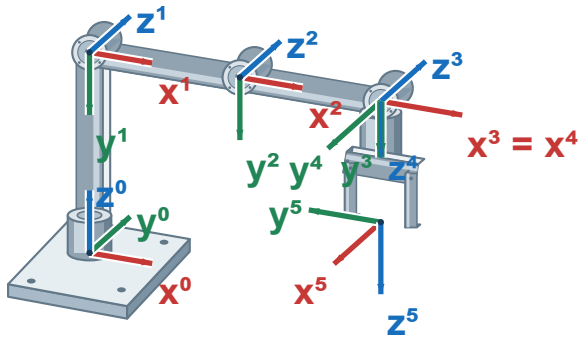
Joint zero

Define the pose where all five joint coordinates are zero.

$$q = [0°\ \ 0°\ \ 0°\ \ 0°\ \ 0°]^T$$

The drawing is a reference pose. Its fifth joint is at 90°, as recorded in the Home column.

Alpha II: drawn reference pose



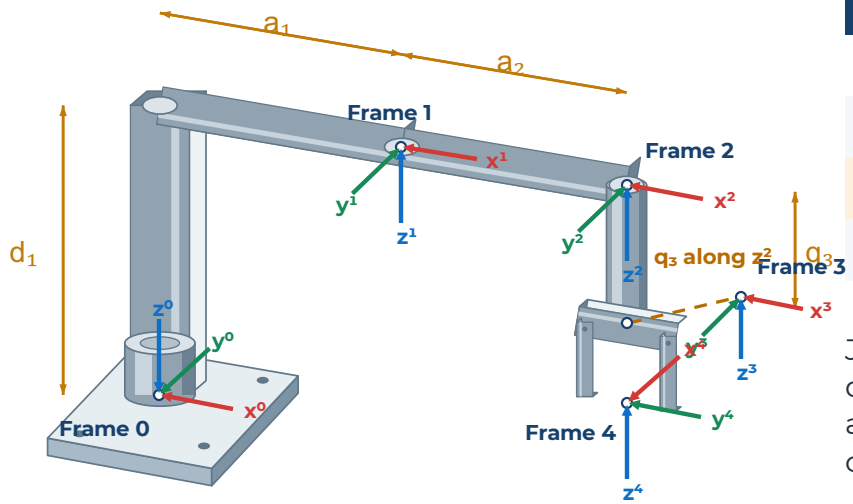
$$q = [0°\ \ 0°\ \ 0°\ \ 0°\ \ 90°]^T$$

The Home column records the reference configuration.
It is separate from the four DH parameters for each joint.

Example 2: Adept One, 4-Axis SCARA

EXAMPLE

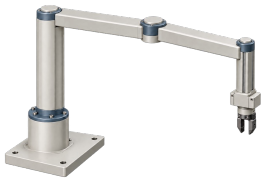
4 joints in an R–R–P–R chain; joint 3 translates



Frame 3 is drawn offset for clarity; its actual origin lies on z^2 after the q_3 translation.

Axis	θ	d	a	α	Home
1	*q1*	*d1*	*a1*	*180°*	*0°*
2	*q2*	*0*	*a2*	*0°*	*0°*
3	*0*	*q3*	*0*	*0°*	*d3*
4	*q4*	*d4*	*0*	*0°*	*−90°*

Joint 3 is prismatic: q_3 is the displacement from Frame 2 to Frame 3 along z^2. Therefore q_3 appears in the d column, while θ_3 is constant.



Pose-matched technical render

PART 3

From DH table to arm equation

Convert neighbouring-frame transforms into forward kinematics

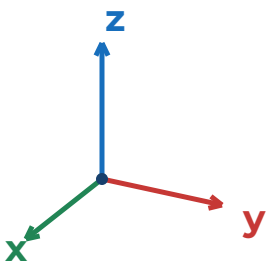
1. Adjacent-frame screw transformation
2. Standard DH transformation matrix
3. Base-to-TCP arm equation and checks

Adjacent Frames: Screw Transformation

CORE

One rigid transform connects two frames fixed to neighbouring links

Frame {i−1}



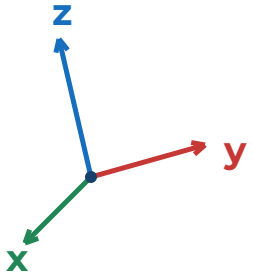
$$A_i(q_i) = {}^{i-1}T_i$$



Equivalent screw axis

Rotation about + translation along one axis

Frame {i}



Standard DH computation

$$A_i = S_z(\theta_i, d_i)\, S_x(\alpha_i, a_i)$$

Each S is one same-axis rotation–translation pair.
Only θ_i or d_i varies with q_i; the remaining DH parameters describe fixed geometry.

The Standard DH Transformation Matrix

CORE

Substitute the four parameters and read off one link transform

$$A_i = \begin{bmatrix} c_{\theta_i} & -s_{\theta_i} c_{\alpha_i} & s_{\theta_i} s_{\alpha_i} & a_i c_{\theta_i} \\ s_{\theta_i} & c_{\theta_i} c_{\alpha_i} & -c_{\theta_i} s_{\alpha_i} & a_i s_{\theta_i} \\ 0 & s_{\alpha_i} & c_{\alpha_i} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Top-left 3×3

— orientation of frame i in frame i−1

Right-hand column

— position of O(i) in frame i−1

Where α appears

— only in columns 2 and 3

Where a and d appear

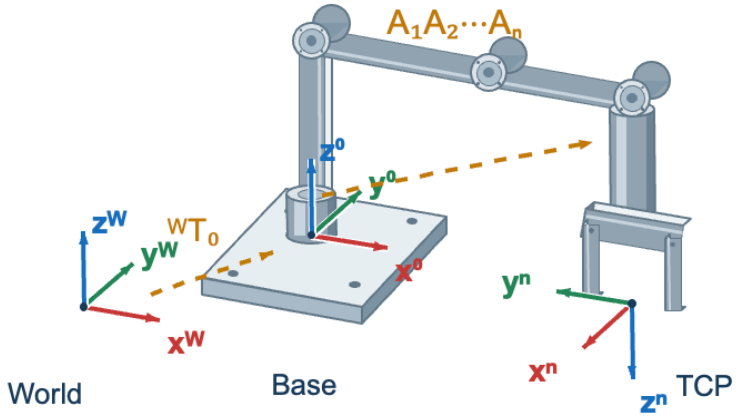
— only in the last column

$$c_\beta = cos\beta, \qquad s_\beta = sin\beta$$

Arm Equation: Base to TCP

CORE

Multiply the adjacent-link transforms, then include the fixed tool transform



$$^{W}T_{TCP} = {}^{W}T_0(A_1A_2 \cdots A_n)^{n}T_{TCP}$$

Mounting position at one end, tool offset at the other. If either is the identity, say why.

Example 1: Alpha II, Individual Link Matrices

EXAMPLE

Each row of the table substituted into A(i)

$$A_1 = \begin{bmatrix} c_1 & 0 & -s_1 & 0 \\ s_1 & 0 & c_1 & 0 \\ 0 & -1 & 0 & d_1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_2 = \begin{bmatrix} c_2 & -s_2 & 0 & a_2c_2 \\ s_2 & c_2 & 0 & a_2s_2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_3 = \begin{bmatrix} c_3 & -s_3 & 0 & a_3c_3 \\ s_3 & c_3 & 0 & a_3s_3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

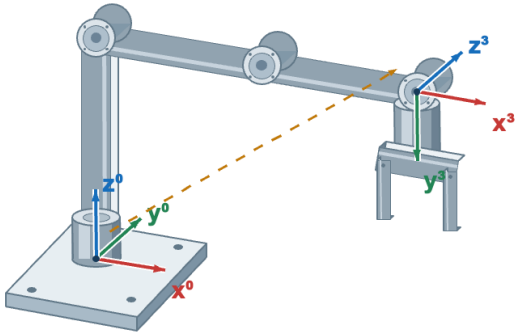
$$A_4 = \begin{bmatrix} c_4 & 0 & -s_4 & 0 \\ s_4 & 0 & c_4 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad A_5 = \begin{bmatrix} c_5 & -s_5 & 0 & 0 \\ s_5 & c_5 & 0 & 0 \\ 0 & 0 & 1 & d_5 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$c_i = \cos\theta_i$ and $s_i = \sin\theta_i$. Each A_i uses the same standard DH convention.

Example 1: Alpha II, Multiplying the Transforms

EXAMPLE

$A_1A_2A_3$: base to wrist



$$T_3 = A_1A_2A_3 = \begin{bmatrix} c_1c_{23} & -c_1s_{23} & -s_1 & c_1(a_2c_2 + a_3c_{23}) \\ s_1c_{23} & -s_1s_{23} & c_1 & s_1(a_2c_2 + a_3c_{23}) \\ -s_{23} & -c_{23} & 0 & d_1 - a_2s_2 - a_3s_{23} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$^0T_5 = A_1A_2A_3A_4A_5$$

Shorthand: $c_{23} = \cos(\theta_2+\theta_3)$, $s_{23} = \sin(\theta_2+\theta_3)$, $c_{234} = \cos(\theta_2+\theta_3+\theta_4)$.

Example 1: Alpha II, Final Arm Equation

EXAMPLE

One 4×4 matrix — orientation in three columns, position in the fourth

$$T_5 = \begin{bmatrix} {}^0R_5 & {}^0\mathbf{p}_5 \\ \mathbf{0}^T & 1 \end{bmatrix}$$

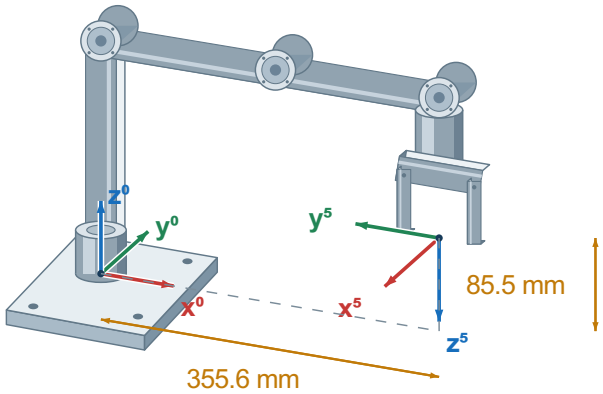
$$T_5 = \begin{bmatrix} c_1c_{234}c_5 + s_1s_5 & -c_1c_{234}s_5 + s_1c_5 & -c_1s_{234} & c_1(a_2c_2 + a_3c_{23} - d_5s_{234}) \\ s_1c_{234}c_5 - c_1s_5 & -s_1c_{234}s_5 - c_1c_5 & -s_1s_{234} & s_1(a_2c_2 + a_3c_{23} - d_5s_{234}) \\ -s_{234}c_5 & s_{234}s_5 & -c_{234} & d_1 - a_2s_2 - a_3s_{23} - d_5c_{234} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The first three columns are the tool axes; the fourth is the position of O_5 — both in frame 0. Frame 5 is the last DH frame; a tool tip elsewhere needs one more fixed transform.

Example 1: Alpha II, Reference Check

EXAMPLE

Substitute a configuration you can see, then check the answer against the drawing



Substitute the defined reference joint coordinates.

Check the expected tool position.

Check the tool-axis directions.

$$R^T R = I, \qquad detR = 1$$

Orthogonality proves R is a rotation; geometry says it is the right one.

At q = (0°, 0°, 0°, 0°, 90°) the arm is horizontal and the tool points straight down:
p_{05} = (355.6, 0, 85.5) mm.

Common Modelling Errors

CORE

Where to look first — these are entry points, not diagnoses

Symptom	What to check
Everything the wrong size	length units — mm against m
Rotates the wrong way	axis direction and joint sign
Right shape, wrong place	base and tool transforms
Error changes with configuration	DH convention, offsets and joint mapping

Two classics: degrees mixed with radians, and Standard DH parameters used in a Modified DH implementation.

PART 4

URDF robot models

Compare a hand-built model with a software robot description

1 Common DH modelling errors
2 URDF joints and imported robot models
3 DH versus URDF and lecture summary

DH and URDF Describe the Same Kinematic Chain

OPTIONAL

The same neighbouring-link transform is organised differently

DH model

$$A_i(q_i) = Rz(q_i + \theta_{oi})\ Tz(d_i)\ Tx(a_i)\ Rx(\alpha_i)$$

- Four parameters define each adjacent frame transform
- The DH convention fixes where the frames are placed
- The joint variable q_i appears inside $A_i(q_i)$

URDF joint

origin xyz / rpy + axis xyz + q_i

- origin gives the fixed parent-to-joint pose at $q_i = 0$
- axis is a direction vector in the joint frame
- the joint state supplies the rotation or translation q_i

Both descriptions must produce the same product ${}^0T_n(q) = A_1(q_1)\ A_2(q_2)\ \ldots\ A_n(q_n)$. URDF axis is not another coordinate frame. It only specifies the direction of joint motion.

IRB2600ID: Standard DH and URDF

OPTIONAL

Row 1: nonzero link length and twist, with every DH value located in the URDF

IRB2600ID-8/2.00 Standard DH

Axis	θ_i	d_i (mm)	a_i (mm)	α_i
1	q_1	445	150	−90°
2	q_2 − 90°	0	900	0°
3	q_3	0	150	−90°
4	q_4	938	0	+90°
5	q_5 + 180°	0	0	+90°
6	q_6	200	0	0°

$\theta_1 = q_1$: joint state about axis [0 0 1]
d_1 = 445 mm: revolute origin z = 0.445
a_1 = 150 mm: fixed origin x = 0.150
α_1 = −90°: fixed origin roll = −1.5708

T = Tz(0.445) Rz(q_1) Tx(0.150) Rx(−π/2)

URDF for row 1 (metres and radians)

```
<joint name="joint1" type="revolute">
  <parent link="base_link"/>
  <child link="joint1_frame"/>
  <origin xyz="0 0 0.445" rpy="0 0 0"/>
  <axis xyz="0 0 1"/>
  <limit lower="-1.57" upper="1.57"
         effort="20" velocity="1.0"/>
</joint>

<joint name="link1_offset" type="fixed">
  <parent link="joint1_frame"/>
  <child link="link1"/>
  <origin xyz="0.150 0 0"
          rpy="-1.57079632679 0 0"/>
</joint>
```

Declare all three links. Limits above are illustrative, not ABB specifications.

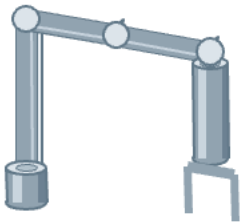
joint1_frame is an intermediate coordinate frame. The fixed joint adds no motor, physical axis or degree of freedom.

DH Modelling: Summary and Checks

CORE

Four outputs you should now be able to produce for your own robot

Frames



Parameter table

θ	d	a	α
.	.	.	.
.	.	.	.
.	.	.	.

Arm equation



Validation



$$^{0}T_{TCP} = A_1 A_2 \cdots A_n \, {}^{n}T_{TCP}$$

1 Assign
consistent frames

2 Determine
the DH parameters

3 Build
the arm equation

4 Check
geometry + software

Can you explain every non-zero entry in your DH table?


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THANK YOU!